

PROJECT EXECUTABLE FILES

Maximum Marks: 3 Marks

Project Description

ComicCraft is a web-based AI application that generates personalized comic stories and illustrations from user-provided prompts. It uses FastAPI for the backend, Google Gemini models for story generation, and Stable Diffusion through Hugging Face Diffusers for comic-style image generation. Users provide a story prompt, character name, setting, tone, and art style. The system generates a panel-by-panel comic, displays a preview, and exports the completed comic as a PDF.

Technology / Project Components

Python

FastAPI and Uvicorn

HTML, CSS and Jinja2 templates

Google Gemini Flash for structured comic panel outlines

Google Gemini Pro for detailed narration and character dialogues

Hugging Face Diffusers with Stable Diffusion for comic illustrations

FPDF for PDF export

Pillow and supporting Python packages

Step 1: Submission Checklist

Step 2: File / Folder Structure

The following structure is based on the project components described in the ComicCraft project document. Actual files should be included in the final submission.

ComicCraft/

■ ■ ■ app/

■ ■ ■ ■ main.py

■ ■ ■ ■ routes.py

■ ■ ■ gemini_flash.py

■ ■ ■ gemini_pro.py

■ ■ ■ image_generator.py

■ ■ ■ layout_builder.py

■ ■ ■ exporters.py

■ ■ ■ templates/

■ ■ ■ ■ index.html

■ ■ ■ ■ comic_preview.html

■ ■ ■ ■ export_success.html

■ ■ ■ static/

■ ■ ■ ■ panels/

■■■■ exports/
■■■■ requirements.txt
■■■■ .env.example
■■■■ README.md

Step 3: Deployment / Access Details

Step 4: Run Instructions

1. Install Python and pip on the system.
2. Open the ComicCraft project folder in a terminal or VS Code.
3. Create a virtual environment using: `python -m venv env`
4. Activate the environment on Windows using: `env\Scripts\activate`
5. Install the required dependencies using: `pip install -r requirements.txt`
6. Configure the required Gemini API key and Hugging Face API token in environment variables or a `.env` file.
7. Start the FastAPI server using: `uvicorn app.main:app --reload`
8. Open the application at `http://127.0.0.1:8000`
9. Use the homepage to enter the story prompt, character name, setting, tone, and art style.
10. Generate the comic, review the panel-by-panel preview, and use the PDF download option to export the comic.
11. FastAPI API documentation can be accessed at `http://127.0.0.1:8000/docs`

Step 5: Known Issues / Limitations

Project Workflow Summary

Milestone 1 – Model selection and architecture

Milestone 2 – Core functionality development

Milestone 3 – routes.py development

Milestone 4 – Frontend development

Milestone 5 – Local deployment and testing

Important Note

This document has been reformatted to follow the uploaded Project Executable Files template. Items such as the actual source-code package, repository URL, deployed URL, demo credentials, test files, and demo video were not present in the supplied project document, so they are marked as not provided/to be attached rather than being invented.

Date	15 March 2026
Team ID	SWTID-2026-7446
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	To be attached
2	README / Setup Guide (instructions to run the project)	Included below
3	requirements.txt / dependency file	To be attached
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	To be attached
6	Deployed application URL (if hosted)	SWTID-2026-7446
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	SWTID-2026-7446
9	Test files and test results	To be attached
10	Demo video or walkthrough (if required)	SWTID-2026-7446

Hosted / Deployed URL	Not provided – project is documented for local deployment.
Login Credentials (Demo)	Not applicable / not provided.
Platform / Hosting Provider	Local deployment using FastAPI and Uvicorn.
Repository Link	Not provided.
Demo Video Link	Not provided.

No	Known Issue / Limitation	Workaround / Status
	The current version focuses on single comic generation and does not include user account management.	Future enhancement: add user profiles and comic libraries.
	AI generation depends on the configured Gemini and Hugging Face services/API credentials.	Configure valid API keys/tokens before running.
	Cloud deployment details are not provided in the source document.	Current documented setup is local deployment using FastAPI/Uvicorn.